Ants on a Stick: Collision Invariance and Exit-Time Bounds

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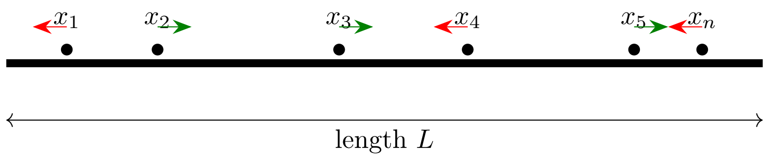

Figure 1: A finite system of identical agents on an interval. Each agent moves at speed v either left or right, reverses direction at collisions, and exits at an endpoint.

Abstract

How long can it take for a group of ants to fall off a stick if every ant moves at the same speed and reverses direction whenever it collides with another ant? The familiar puzzle has a deceptively simple answer: if the ants are indistinguishable, a collision can be treated as two ants passing through one another and exchanging labels. Thus the last exit time depends only on the farthest directed distance to an endpoint.

This paper develops that observation into a deterministic, algorithmic, and distributional analysis of an interacting-agent system on an interval. We formalize the collision-invariance principle, characterize the full multiset of exit times, prove sharp extremal completion-time bounds, give optimal algorithms for completion time and ordered exit times, and count the number of possible physical collisions. Under iid uniform initial positions with a direction vector independent of the positions, the exit process reduces to order statistics, yielding exact formulas for the completion-time distribution, expectation, variance, quantiles, and the number of agents remaining at a given time. We also identify a direction-law invariance under uniform positions, derive the expected completion time for a general directed-distance distribution, and formalize the almost-sure limit of repeated-trial average completion times. A final section interprets the model as a stylized time-to-clear problem: not a literal financial-market model, but a compact example of how an apparently interaction-heavy system can collapse into a tractable bottleneck variable.

1 The Ants-on-a-Stick Problem

The classic ants-on-a-stick puzzle asks for the maximum time required for a group of ants to fall from a stick when all ants move at the same speed and reverse direction at collisions. A standard version asks about ten ants on a two-foot stick moving at one foot per minute. At first glance, the problem appears to require tracking a sequence of pairwise collisions. The key observation is that if the ants are identical, then a collision between two ants is equivalent to the ants passing through each other while exchanging labels. This is the familiar solution to the recreational puzzle, but it also suggests a more general modeling principle: some one-dimensional interacting systems have unlabeled observables that are invariant under collision dynamics.[5;4]

The goal of this paper is not to rediscover the classic trick. Instead, the goal is to build a complete exit-time framework around it. The analysis proceeds in four layers. First, we formulate a deterministic model and prove the collision-invariance lemma. Second, we characterize not only the final completion time but the entire unordered exit-time process. Third, we show that the completion time can be computed by an optimal linear-time algorithm even though the physical collision count may be quadratic. Finally, we impose random initial conditions and derive exact distributional, asymptotic, and risk-style quantities.

For the motivating classroom version, the stick length is

$$L = 2 \text{ feet}$$

and the common speed is

$$v = 1 \text{ foot/minute.}$$

The sharp upper bound on completion time is therefore two minutes. The sections below explain why this bound is independent of the number of ants and why the same model yields richer stochastic information when initial positions and directions are randomized.

2 Deterministic Model

Consider the interval

$$[0, L]$$

with length $L > 0$. There are $n \geq 1$ point agents, called ants, initially located at positions

$$x_1, x_2, \ldots, x_n \in [0, L].$$

Each ant moves at common speed $v > 0$ and has initial direction

$$d_i \in \{-1, +1\},$$

where $d_i = -1$ denotes motion to the left and $d_i = +1$ denotes motion to the right. An ant moving from the interior exits when it reaches either endpoint. For the endpoint-inclusive extension of the model, an ant initialized exactly at an endpoint and directed inward is regarded as still on the stick at time zero, whereas an ant initialized there and directed outward exits immediately. When two ants collide, they instantaneously reverse directions and continue moving at speed v.

The main assumptions are:

- ants are modeled as point particles;
- all ants have the same speed v;
- collisions are instantaneous and elastic in the sense of direction reversal;
- labels are not relevant to the system completion time;
- simultaneous collisions are excluded in the base model, or handled as limits of configurations with distinct collision times.

The equal-speed assumption is essential. If agents have different speeds and retain their speeds after collision, the simple label-swap equivalence can fail. The present paper studies the equal-speed case because that is the setting in which the collision process admits an exact reduction.

3 Collision Invariance

The mathematical center of the problem is the observation that for identical ants, a collision can be removed by relabeling.

Lemma 1 (Collision invariance)**.** *Consider two identical ants moving at equal speed on a line. The evolution in which the ants collide and reverse directions produces the same unordered set of future positions as the evolution in which the ants pass through each other and exchange labels.*

Proof. Immediately before a collision, one ant approaches from the left and moves right, while the other approaches from the right and moves left. In the physical collision model, both ants reverse direction at the meeting point. In the pass-through model, both ants continue moving through the meeting point. Since the ants have equal speed and are identical, the set of occupied positions after the meeting is the same in both descriptions. The only difference is the label assigned to each trajectory. If labels are ignored, collision-and-reversal and pass-through motion are equivalent. □

This lemma is local, but applying it at every collision removes the entire interacting structure. The physical system can therefore be analyzed through a noninteracting pass-through system, provided the quantity of interest depends only on the set of occupied positions or exit times and not on the identity of a particular named ant.

4 Exit Times and Extremal Bounds

For each ant define its directed distance to the endpoint it initially faces:

$$\delta_i = \begin{cases} x_i, & d_i = -1, \\ L - x_i, & d_i = +1. \end{cases} \tag{1}$$

In the pass-through system, ant i exits after time δ_i/v.

Theorem 1 (Exit-time multiset)**.** *The unordered multiset of physical exit times under collision-and-reversal dynamics is*

$$\{\tau_1, \dots, \tau_n\} = \left\{ \frac{\delta_1}{v}, \dots, \frac{\delta_n}{v} \right\}.$$

Consequently, if $\delta_{(1)} \le \delta_{(2)} \le \cdots \le \delta_{(n)}$ are the ordered directed exit distances, then the k-th exit time is

$$T_{(k)} = \frac{\delta_{(k)}}{v}.$$

In particular, the completion time is

$$T = T_{(n)} = \frac{1}{v} \max_{1 \le i \le n} \delta_i. \tag{2}$$

Proof. By repeated application of the collision-invariance lemma, the interacting collision model has the same unordered set of occupied positions as the pass-through model at every time before exit. In the pass-through model, each trajectory moves in its initial direction until it reaches the corresponding endpoint, so its exit time is δ_i/v. Since collisions only exchange labels, the physical system has the same unordered multiset of exit times. Ordering those times gives $T_{(k)} = \delta_{(k)}/v$, and the last exit time is the maximum. $\square$

Maximum completion time

Since $0 \le \delta_i \le L$ for every i, equation (2) gives

$$T \le \frac{L}{v}. \tag{3}$$

The sharp upper bound is therefore

$$\sup T = \frac{L}{v}. \tag{4}$$

Under the endpoint-inclusive convention of Section 2, an ant may begin at $x = 0$ facing right or at $x = L$ facing left, so the bound is attained and

$$T_{\max} = \frac{L}{v}.$$

If all initial positions are restricted to the open interval $(0, L)$, then configurations can approach this value arbitrarily closely but cannot attain it, so L/v is the supremum rather than a maximum.

For the two-foot, one-foot-per-minute version,

$$T_{\max} = \frac{2}{1} = 2 \text{ minutes} = 120 \text{ seconds}.$$

Minimum completion time

The minimum likewise depends on the admissible initial configurations. Under the endpoint-inclusive convention, if coincident initial positions at the boundary are permitted, all ants may be initialized at endpoints facing outward and therefore exit at time zero, so

$$T_{\min} = 0.$$

If initial positions must lie strictly in $(0, L)$, then $T = 0$ is not attained, but the infimum is still 0.

Exercise 1 (Minimum completion time)**.** Assume all ants begin at distinct points in the open interval $(0, L)$. Show that the completion time has no positive lower bound. In particular, for every $\varepsilon > 0$, construct a configuration of n ants with $T < \varepsilon$.

Solution. Choose distinct positions $x_i \in (0, \min\{v\varepsilon, L\})$ and direct every ant to the left. Then $\delta_i = x_i < v\varepsilon$ for all i, so

$$T = \max_i \frac{x_i}{v} < \varepsilon.$$

Therefore $\inf T = 0$, although no open-interval configuration attains $T = 0$.

5 Algorithms and Complexity

The direct physical simulation of the ant system requires tracking collisions and exits. The collision-invariance theorem eliminates this burden for the completion-time problem. The following algorithm computes the last exit time directly from the initial configuration. As in the mathematical model, the listing assumes equally sized position and direction arrays and valid inputs with $L > 0$, $v > 0$, $x_i \in [0, L]$, and $d_i \in \{-1, +1\}$; the accompanying source file additionally performs explicit input validation.

Listing 1: Linear-time computation of completion time.

```
def completion_time(positions, directions, L, v):
  ans = 0.0
  for x, d in zip(positions, directions):
    delta = x if d == -1 else L - x
    ans = max(ans, delta / v)
  return ans
```

The runtime is $O(n)$ because the algorithm performs one constant-time computation per ant. This is also optimal in the standard input model; the argument follows the usual input-reading lower-bound logic used in asymptotic algorithm analysis.[1]

Theorem 2 (Optimality for completion time)**.** *Any deterministic algorithm that computes the completion time for arbitrary inputs must inspect every ant in the worst case. Therefore the completion time problem has a lower bound $\Omega(n)$, and the one-pass algorithm above is $\Theta(n)$.*

Proof. Suppose an algorithm does not inspect ant j. Construct two inputs that are identical on every inspected ant but differ in the uninspected ant j: in one input, ant j has a small directed exit distance; in the other, it has directed exit distance close to L. These two inputs can have different completion times while producing the same information to the algorithm. Thus any correct algorithm must inspect every ant in the worst case. The one-pass maximum computation matches this lower bound. □

For the full ordered exit sequence, one computes all directed exit times δ_i/v and sorts them. This takes $O(n \log n)$ time by comparison sorting. The matching lower bound follows by reduction from comparison sorting: fix $v = 1$, take every ant to move left, and encode arbitrary distinct keys $a_i \in (0, L)$ as positions $x_i = a_i$. The ordered exit-time sequence is then exactly the sorted key sequence. Hence any comparison-based algorithm that outputs all exit times in order requires $\Omega(n \log n)$ comparisons in the worst case, and the ordered exit sequence is computable in $\Theta(n \log n)$ time.[1]

This contrast is the main algorithmic lesson: the full ordered process requires sorting, while the final completion time requires only a maximum.

6 Counting Collisions

Collisions do not affect the exit-time multiset, but they may still be numerous. Assume the initial positions are distinct and sorted so that

$$x_1 < x_2 < \cdots < x_n.$$

In the pass-through picture, a physical collision occurs exactly when a right-moving trajectory from the left crosses a left-moving trajectory from the right. Therefore the collision count is

$$C = \#\{(i, j) : i < j,\ d_i = +1,\ d_j = -1\}. \tag{5}$$

This is an inversion statistic on the direction sequence.

Theorem 3 (Worst-case collision count)**.** *The maximum possible number of physical collisions is*

$$C_{\max} = \left\lfloor \frac{n^2}{4} \right\rfloor .$$

Proof. Let r be the number of right-moving ants and ℓ the number of left-moving ants, so $r + \ell = n$. Every collision corresponds to an inversion in the direction sequence, so there are at most $r\ell$ collisions. Equality is attained when all r right-moving ants begin to the left of all ℓ left-moving ants, in which case every right-left pair crosses exactly once. The product $r\ell = r(n - r)$ is maximized when the two groups are as balanced as possible, giving $\lfloor n^2/4 \rfloor$. □

If directions are independent fair coin flips, then for each ordered pair $i < j$ the event $(d_i, d_j) = (+1, -1)$ has probability $1/4$. By linearity of expectation,

$$\mathbb{E}[C] = \binom{n}{2}\frac{1}{4} = \frac{n(n-1)}{8}. \tag{6}$$

Thus the physical system can contain $\Theta(n^2)$ collisions even though the completion time is computable in $\Theta(n)$ time.

7 Random Initial Conditions

The deterministic theorem becomes more informative when initial configurations are random. As a canonical model, assume $n \geq 1$ and

$$X_i \sim \text{Uniform}(0, L), \qquad \mathbb{P}(D_i = -1) = \mathbb{P}(D_i = +1) = \frac{1}{2},$$

with positions and directions mutually independent across ants. The directed exit distance is

$$\Delta_i = X_i \quad \text{if } D_i = -1, \qquad \Delta_i = L - X_i \quad \text{if } D_i = +1.$$

The fair-direction assumption is natural for the canonical experiment, but it is stronger than is needed for the completion-time law.

Theorem 4 (Direction-law invariance under uniform positions)**.** *Let $X_1, \ldots, X_n$ be iid* Uniform$(0, L)$*, and let the direction vector*

$$D = (D_1, \ldots, D_n) \in \{-1, +1\}^n$$

have an arbitrary distribution that is independent of

$$X = (X_1, \ldots, X_n).$$

The components of D need not be fair or mutually independent. Then the directed exit distances $\Delta_1, \ldots, \Delta_n$ are iid Uniform$(0, L)$*.*

Proof. Condition on any fixed direction vector $D = d$. For each coordinate, the map from X_i to Δ_i is either the identity $x \mapsto x$ or the reflection $x \mapsto L - x$. Both maps preserve the uniform distribution on $[0, L]$. Because the X_i are independent, the conditional joint law of $(\Delta_1, \ldots, \Delta_n)$ given $D = d$ is the product of n uniform laws on $[0, L]$. Crucially, this conditional law does not depend on d. Averaging over the arbitrary distribution of D therefore leaves the same product law unaltered. Hence the directed exit distances are iid Uniform$(0, L)$. □

Thus the exit-time process is governed by order statistics of independent uniform random variables whenever positions are iid uniform and the direction vector is independent of the positions.[7;3;2] Fair, independent directions remain necessary for the collision-count expectation derived in the preceding section, but not for the completion-time distribution.

The k-th exit time

Let

$$0 \le \Delta_{(1)} \le \Delta_{(2)} \le \cdots \le \Delta_{(n)} \le L$$

be the ordered directed exit distances. Then

$$\frac{\Delta_{(k)}}{L} \sim \mathrm{Beta}(k, n+1-k),$$

and hence

$$\frac{vT_{(k)}}{L} \sim \mathrm{Beta}(k, n+1-k). \tag{7}$$

In particular,

$$\mathbb{E}[T_{(k)}] = \frac{k}{n+1}\frac{L}{v}.$$

The expected first exit time is therefore

$$\mathbb{E}[T_{(1)}] = \frac{1}{n+1}\frac{L}{v},$$

while the expected last exit time is

$$\mathbb{E}[T_{(n)}] = \frac{n}{n+1}\frac{L}{v}.$$

Completion-time distribution

Let $T = T_{(n)}$. For $0 \le t \le L/v$,

$$\mathbb{P}(T \le t) = \mathbb{P}\left(\Delta_1 \le vt, \ldots, \Delta_n \le vt\right) \tag{8}$$

$$= \left(\frac{vt}{L}\right)^n. \tag{9}$$

The density is therefore

$$f_T(t) = n\left(\frac{v}{L}\right)^n t^{n-1}, \qquad 0 \le t \le \frac{L}{v}. \tag{10}$$

For every positive integer k, the moments are

$$\mathbb{E}[T^k] = \frac{n}{n+k}\left(\frac{L}{v}\right)^k, \tag{11}$$

so

$$\mathbb{E}[T] = \frac{n}{n+1}\frac{L}{v}, \tag{12}$$

and

$$\mathrm{Var}(T) = \frac{n}{(n+1)^2(n+2)}\left(\frac{L}{v}\right)^2. \tag{13}$$

For the two-foot, one-foot-per-minute model,

$$\mathbb{E}[T] = \frac{2n}{n+1} \text{ minutes.}$$

For example,

$$n = 10: \quad \mathbb{E}[T] = \frac{20}{11} \approx 1.818 \text{ minutes,}$$

while

$$n = 50: \quad \mathbb{E}[T] = \frac{100}{51} \approx 1.961 \text{ minutes.}$$

As the number of ants grows, the expected completion time approaches the worst-case time of two minutes.

Configuration-space mean and repeated trials

The expectation above can also be interpreted directly as an average over the entire continuous configuration space. Under the canonical fair-direction model, let

$$\Omega_n = [0, L]^n \times \{-1, +1\}^n,$$

equipped with normalized n-dimensional Lebesgue measure on the position cube and the uniform probability measure on the 2^n direction vectors. For a configuration

$$\omega = (x_1, \dots, x_n, d_1, \dots, d_n),$$

let $T(\omega)$ denote the time at which the last ant exits. The maximum in

$$T(\omega) = \frac{1}{v} \max_i \Delta_i(\omega)$$

is taken across ants *within a single configuration*; the expectation $\mathbb{E}[T]$ averages this configuration-level completion time over the probability space of initial configurations.

Theorem 5 (Configuration-space mean and infinite-trial limit)**.** *Under iid uniform initial positions and any direction law independent of those positions,*

$$\mathbb{E}[T] = \frac{n}{n+1} \frac{L}{v}. \tag{14}$$

If $T_1, T_2, \dots$ are completion times from independent repetitions of the same experiment with fixed n, L, and v, then

$$\overline{T}_m = \frac{1}{m} \sum_{j=1}^{m} T_j \longrightarrow \frac{n}{n+1} \frac{L}{v} \qquad \textit{almost surely as } m \to \infty. \tag{15}$$

Proof. For the canonical fair-direction measure, the expectation may be written as

$$\mathbb{E}[T] = \frac{1}{2^n L^n v} \sum_{d \in \{-1,+1\}^n} \int_{[0,L]^n} \max_{1 \le i \le n} \Delta_i(x_i, d_i)\, dx_1 \cdots dx_n.$$

For every fixed d, the coordinate-wise map $x_i \mapsto \Delta_i$ is a composition of identities and reflections of $[0, L]$, hence is measure preserving. Each integral in the direction sum is therefore the same as the integral of the maximum coordinate over the uniform cube. Equivalently, if

$$M = \max_{1 \le i \le n} \Delta_i,$$

then

$$\mathbb{P}(M \le y) = \left(\frac{y}{L}\right)^n, \qquad 0 \le y \le L.$$

Using the tail-integral identity for a nonnegative bounded random variable,

$$\begin{aligned}
\mathbb{E}[T] &= \frac{1}{v}\mathbb{E}[M] && (16)\\
&= \frac{1}{v} \int_0^L \mathbb{P}(M > y)\, dy && (17)\\
&= \frac{1}{v} \int_0^L \left[1 - \left(\frac{y}{L}\right)^n\right] dy && (18)\\
&= \frac{n}{n+1} \frac{L}{v}. && (19)
\end{aligned}$$

The direction-law invariance theorem shows that the same mean holds for any direction distribution independent of the iid uniform positions. Finally, $0 \le T_j \le L/v$, so $\mathbb{E}[|T_j|] < \infty$. The strong law of large numbers therefore gives equation (15).[3;7] □

For a finite Monte Carlo sample of m independent trials,

$$\operatorname{Var}(\overline{T}_m) = \frac{1}{m} \operatorname{Var}(T) = \frac{n}{m(n+1)^2(n+2)} \left(\frac{L}{v}\right)^2. \tag{20}$$

Thus repeated simulation estimates the configuration-space mean with sampling variance that decays at rate $1/m$.

8 Asymptotics and Ants Remaining

Let

$$M_n = \frac{vT}{L}.$$

Under the uniform model, M_n is the maximum of n independent Uniform$(0,1)$ random variables. Therefore

$$M_n \to 1$$

in probability as $n \to \infty$. More sharply, for $0 \le x \le n$,

$$\mathbb{P}\left(n(1-M_n) > x\right) = \left(1 - \frac{x}{n}\right)^n.$$

Hence, for every fixed $x \ge 0$ as $n \to \infty$,

$$\mathbb{P}\left(n(1-M_n) > x\right) \to e^{-x},$$

so

$$n\left(1 - \frac{vT}{L}\right) \Rightarrow \text{Exponential}(1). \tag{21}$$

Thus the gap between the completion time and the worst-case bound is typically of order $1/n$.

The same model also gives the number of ants remaining at time t. Define

$$R(t) = \#\{i : \Delta_i > vt\}.$$

For $0 \le t \le L/v$,

$$R(t) \sim \text{Binomial}\left(n, 1 - \frac{vt}{L}\right). \tag{22}$$

Thus

$$\mathbb{E}[R(t)] = n\left(1 - \frac{vt}{L}\right), \tag{23}$$

and

$$\mathrm{Var}(R(t)) = n\left(1 - \frac{vt}{L}\right)\left(\frac{vt}{L}\right). \tag{24}$$

This gives a direct theoretical explanation for an interactive simulation counter showing how many ants remain on the stick at each time.

9 General Initial Distributions

The uniform model is only one special case. Suppose the initial positions $X_1, \ldots, X_n$ are iid with continuous cumulative distribution function F on $[0, L]$. Suppose also that the directions $D_1, \ldots, D_n$ are iid, independent of the positions, and satisfy

$$\mathbb{P}(D_i = -1) = p, \qquad \mathbb{P}(D_i = +1) = 1 - p.$$

Then the directed exit distance Δ_i has CDF

$$G(y) = pF(y) + (1-p)\left[1 - F(L-y)\right], \qquad 0 \le y \le L. \tag{25}$$

Indeed, the first term corresponds to left-moving ants with $X_i \le y$, while the second corresponds to right-moving ants with $L - X_i \le y$, or equivalently $X_i \ge L - y$. Continuity of F removes any distinction between strict and weak inequalities at the threshold.

Theorem 6 (General completion-time distribution)**.** *Under the assumptions above, the completion time satisfies*

$$\mathbb{P}(T \le t) = G(vt)^n, \qquad 0 \le t \le \frac{L}{v}, \tag{26}$$

where G is defined in equation (25).

Proof. By the exit-time theorem, $T \le t$ if and only if every directed exit distance satisfies $\Delta_i \le vt$. Since the Δ_i are iid with CDF G,

$$\mathbb{P}(T \le t) = \prod_{i=1}^{n} \mathbb{P}(\Delta_i \le vt) = G(vt)^n.$$

□

Theorem 7 (General expected completion time)**.** *Under the same assumptions,*

$$\mathbb{E}[T] = \frac{1}{v} \int_0^L \left[1 - G(y)^n\right] \, dy. \tag{27}$$

Proof. The completion time is nonnegative and bounded above by L/v. Therefore the tail-integral identity gives

$$\mathbb{E}[T] = \int_0^{L/v} \mathbb{P}(T > t) \, dt.$$

Using the preceding theorem,

$$\mathbb{P}(T > t) = 1 - G(vt)^n.$$

The change of variables $y = vt$ then yields

$$\mathbb{E}[T] = \frac{1}{v} \int_0^L \left[1 - G(y)^n\right] \, dy.$$

□

For the uniform position law $F(y) = y/L$, one has

$$1 - F(L - y) = \frac{y}{L},$$

and therefore

$$G(y) = \frac{y}{L}$$

for every $p \in [0, 1]$, not only for $p = 1/2$. Equation (27) then reduces to

$$\mathbb{E}[T] = \frac{n}{n+1} \frac{L}{v}.$$

This general theorem separates the collision dynamics from the input distribution. Once the directed-distance CDF is known, both the completion-time law and its expectation follow directly.

10 Simulation and Verification

The theory suggests two computational approaches. The first simulates the physical system directly by finding collision and exit events. The second ignores collisions and computes the directed exit times. The second method is not an approximation; it is an exact reduction. For compactness, the listing below assumes inputs satisfy the model domain stated in Section 2; the accompanying source file performs explicit validation.

Listing 2: Reduced JavaScript computation for completion time and exit sequence.

```
function directedExitTimes(positions, directions, L, v) {
 return positions.map((x, i) => {
  const delta = directions[i] === -1 ? x : L - x;
  return delta / v;
 });
}

function completionTime(positions, directions, L, v) {
 const times = directedExitTimes(positions, directions, L, v);
 let ans = 0;
 for (const t of times) ans = Math.max(ans, t);
 return ans;
}

function orderedExitTimes(positions, directions, L, v) {
 return directedExitTimes(positions, directions, L, v)
  .sort((a, b) => a - b);
}
```

A direct simulator is still useful for visualization. It shows the apparent complexity of collisions, while the reduced computation provides a correctness oracle. A direct collision simulator and the reduced pass-through computation should agree on the unordered exit-time multiset and the final completion time. If they do not, then the simulator likely contains an error in collision handling, event ordering, or endpoint processing.

For Monte Carlo validation under the uniform model, one may generate independent configurations, compute the corresponding completion times $T_1, \ldots, T_m$, and compare the empirical CDF with

$$F_T(t) = \left(\frac{vt}{L}\right)^n.$$

The sample mean

$$\overline{T}_m = \frac{1}{m}\sum_{j=1}^{m} T_j$$

provides a second diagnostic: by equation (15), it converges almost surely to the exact configuration-space mean

$$\frac{n}{n+1}\frac{L}{v}.$$

Simulation is not needed to prove either result, but it provides an independent computational check of the analytic law.[6;7]

An interactive implementation may use the same reduction to visualize deterministic inputs or stochastic realizations without numerically resolving collisions. Initial positions and directions may be supplied directly or sampled from a stated probability law, while the exit times are computed from the corresponding directed distances. A direct collision simulation may be rendered alongside the reduced computation as an independent visual check, and deterministic benchmark configurations may be shown alongside random trials provided that they remain logically separate from the stochastic sample. These implementation choices illustrate the theory but do not alter the underlying model or its analytic results.

11 Risk-Style Interpretation and Applications

The ant system is not a literal financial-market model. It contains no order book, price process, liquidity constraint, strategic behavior, or stochastic volatility. Still, it is a useful stylized model of a time-to-clear system: completion occurs only when the slowest directed component exits. The relevant variable is therefore an extreme value,

$$T = \max_i \frac{\delta_i}{v}.$$

This makes the problem a compact example of bottleneck risk.

Under the uniform model, for $q \in (0,1)$ the q-quantile of completion time is

$$Q_q(T) = \frac{L}{v} q^{1/n}. \tag{28}$$

A VaR-style time-to-clear measure may therefore be written as

$$\mathrm{VaR}_q(T) = \frac{L}{v} q^{1/n}. \tag{29}$$

The corresponding expected tail completion time is

$$\mathbb{E}\left[T \mid T > Q_q(T)\right] = \frac{L}{v} \cdot \frac{n}{n+1} \cdot \frac{1 - q^{(n+1)/n}}{1-q}. \tag{30}$$

This is analogous in spirit to expected shortfall: it averages completion times conditional on being in the upper tail.[8]

The defensible applications are methodological rather than literal. The analysis illustrates:

- **Algorithmic simplification:** a collision-heavy physical process reduces to a one-pass maximum computation.
- **Simulation verification:** the reduced formula provides an exact oracle for testing event-driven collision simulators.
- **Bottleneck modeling:** system completion is governed by the slowest directed exit distance.
- **Time-to-clear risk:** exact quantiles and tail expectations can be derived for a stylized completion-time variable.
- **Interactive education:** the problem gives a visual example of how invariance can replace brute-force simulation.

The limitations are equally important. The reduction requires equal speeds, point particles, instantaneous collisions, and unlabeled observables. It does not model named-agent trajectories, finite body lengths, strategic behavior, queueing, or real financial market microstructure.

12 Conclusion

The ants-on-a-stick puzzle appears to be a many-body collision problem, but its exit-time structure is governed by a simple invariant. Collision-and-reversal is equivalent, for unlabeled exit times, to pass-through motion with relabeling. Consequently, the full exit-time multiset is exactly

$$\left\{ \frac{\delta_1}{v}, \ldots, \frac{\delta_n}{v} \right\},$$

and the completion time is

$$T = \frac{1}{v} \max_i \delta_i.$$

The sharp worst-case bound is L/v: it is attained under the endpoint-inclusive convention and is otherwise the supremum for open-interval initial positions. The open-interval minimum is not attained but has infimum zero.

The algorithmic consequences are sharp: completion time is computable in optimal $\Theta(n)$ time, even though the physical collision count can be $\Theta(n^2)$. Under random initial conditions, the exit process becomes an order-statistic problem. This yields exact distributions, expectations, variances, quantiles, asymptotic laws, and remaining-agent counts. More generally, the directed-distance CDF G determines

$$\mathbb{E}[T] = \frac{1}{v}\int_0^L \left[1 - G(y)^n\right]\, dy,$$

while for iid uniform positions the repeated-trial average converges almost surely to

$$\frac{n}{n+1}\frac{L}{v}.$$

The result is therefore more than a puzzle trick. It is a compact example of how an interaction-heavy system can collapse into a deterministic metric, an optimal algorithm, and an exact stochastic completion-time model.

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